Yun-Ta Hsieh

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Education

University of Pennsylvania, M.A.S. in Computer Science — Present

University of Michigan, M.S. in Digital and Material Technologies — Expected Dec. 2026

Huazhong University of Science and Technology, B.Arch. — Jun. 2024

Core Courses

- **Mathematics:** Calculus, Linear Algebra, Probability and Statistics, Discrete Mathematics
- **Computer Science:** Software Development and Design, Data Structures and Algorithms, Computer Network, Principles of Programming Languages, Survey of Artificial Intelligence, Operating Systems, Database Systems, Computer Architecture, Information Security, Network Programming, Knowledge Systems

Publications & Manuscripts

QuantVLA: Scale-Calibrated Post-Training Quantization for Vision-Language-Action Models
Jingxuan Zhang*, ***Yun-Ta Hsieh****, Zhongwei Wan, Haokun Lin, Xin Wang, Ziqi Wang, Yingtie Lei, Mi Zhang
CVPR 2026

OS-Omni: A Cross-Platform Benchmark for Generalist Computer-Using Agents
Hui Shen*, ***Yun-Ta Hsieh****, Jianing Ma*, Ziyuan Liu*, Qi Han*, Xiuqi Xu*, Yanheng Shang*, Chenqi Yin, Zesen Zhao, Boyuan Zheng, Jisen Li, Xiaoxia Wu, Ruoyan Zhang, Dongfu Jiang, Ruiyao Liu, Jing Xiong, He Xiao, Nan Zheng, Hangxiang Chen, Shengyang Tao, Deyang Luorong, Jingxuan Zhang, Chaofan Tao, Yuren Wang, Jiaqi Mo, Hei Ting Una Chan, Sicheng Chen, Kai Zhang, Mengkang Hu, Zhongwei Wan, Xin Wang, Chuanyang Zheng, Ziheng Zhang, Hengyuan Zhang, Zunhai Su, Kanzhi Cheng, Jiawei Lu, Yifan Zhang, Shen Yan, Ben Athiwaratkun, Qiushi Sun, Mi Zhang, Ping Luo, Wenhu Chen, Ngai Wong
Submitted to NeurIPS 2026; under review.

Speculative Decoding for Multimodal Models: A Survey
Yuyue Zhang, Y. Wang, ***Yun-Ta Hsieh***, Xin Wang, Peng Zhang, Z. Yang, J. Ma, Z. Zhao, et al.
TMLR 2026

MMFormalizer: Multimodal Autoformalization in the Wild
Jing Xiong, Qi Han, ***Yun-Ta Hsieh***, Hui Shen, Hongsheng Xin, Chaofan Tao, Chenyu Zhao, et al.
arXiv preprint, 2026.

MMSpec: Benchmarking Speculative Decoding for Vision-Language Models
Hui Shen, Xin Wang, Peng Zhang, ***Yun-Ta Hsieh***, Qi Han, Zhongwei Wan, Zhe Zhang, Jingxuan Zhang, et al.
arXiv preprint, 2026.

Prefilling-dLLM: Predictive Prefilling for Long-Context Inference in Diffusion Language Models
Jing Xiong, Qi Han, Shansan Gong, ***Yun-Ta Hsieh***, Chengyue Wu, Chaofan Tao, Chenyang Zhao, Ngai Wong
EMNLP 2026

MathGen: Revealing the Illusion of Mathematical Competence through Text-to-Image Generation
R. Liu, Hui Shen, Peng Zhang, ***Yun-Ta Hsieh***, Yuyue Zhang, J. Xu, S. Chen, et al.
ECCV 2026 Workshop

PhyX: Does Your Model Have the "Wits" for Physical Reasoning?

Hui Shen, Taiqiang Wu, Qi Han, ***Yun-Ta Hsieh***, Jizhou Wang, Yuyue Zhang, Yuxin Cheng, et al.
arXiv preprint, 2025.

SkillEvolBench: Benchmarking the Evolution from Episodic Experience to Procedural Skills

Yingtie Lei, Zhongwei Wan, Jiankun Zhang, Samiul Alam, Zixuan Zhong, Peizhou Huang, Xin Wang, Jingxuan Zhang, Donghao Zhou, ***Yun-Ta Hsieh***, Zhihao Dou, Hui Shen, Yan Xu, Dimitrios Dimitriadis, Tuo Zhang, Mi Zhang
Submitted to NeurIPS 2026; under review.

*Equal contribution.

Research Interests

Fields: Efficient AI, ML Systems, Multimodal AI, VLA Models

Research Experience

Research Assistant, Advisor: Prof. Mi Zhang — Aug. 2025 – Present
AIoT and Machine Learning Systems Lab, Department of Computer Science and Engineering
The Ohio State University, Columbus, OH

- Co-developed **QuantVLA**, a post-training quantization framework for vision-language-action models, identifying quantization-induced attention temperature and residual energy drift and developing scale-calibration methods for stable low-bit inference; achieved 97.6% average success on LIBERO with OpenPI π0.5 and GR00T N1.5 while reducing the memory footprint of π0.5 from 4.27 GB to 1.28 GB.
- Contributed to **MMSpec**, a speculative decoding framework for vision-language models, investigating low draft acceptance caused by weak visual understanding and cross-modal alignment in lightweight VLM drafters.

Research Assistant, Advisor: Prof. Ngai Wong — April. 2025 – Present
Department of Electrical and Computer Engineering
The University of Hong Kong, Hong Kong

- Contributed to **PhyX**, a large-scale benchmark for physics-grounded reasoning in visual scenarios, comprising 3,000 multimodal questions across six reasoning types, 25 subdomains, and six core physics domains.
- Co-developed **OS-Omni**, a cross-platform evaluation framework for generalist computer-using agents, leading the Windows environment, task construction, and evaluation pipeline.

Industry Experience

Research Intern, **TouchstoneAI** — Jun. 2026 – Aug. 2026
AI Startup

- Designed and implemented an end-to-end evaluation framework for LLM-based coding agents, encompassing game-development task formulation and automated evaluation pipeline construction.

Skills

Programming Languages: Python, C/C++, SQL, HTML/CSS

Tools and Frameworks: Git, PyTorch, LaTeX, Docker, Markdown